

Band-Limitedness of the Hardy Z-Function in the Phase Variable

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Abstract

Let ζ denote the Riemann zeta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z-function, and $\theta(t) = \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ the Riemann–Siegel theta function. Set $c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi)$; for any constant $c > c_*$, define the shifted phase $\Theta(t) := \theta(t) + ct$. Then $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$ and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing C^∞ bijection. The unitary inverse phase pullback

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \quad u \in \mathbb{R}, \quad (1)$$

with the principal (positive) square root of the strictly positive real quantity $\Theta'(\Theta^{-1}(u))$, is a real-valued function on $\mathbb{R}$ that admits a Cramér spectral representation $Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi)$ with a finite complex Borel measure $d\Phi$ supported in $[-1, 1]$, and whose power spectral density $S(\mu) := \lim_{T \rightarrow \infty} 2\pi |K_T(\mu)|^2 / \Theta(T)$ vanishes for $|\mu| > 1$. From the spectral representation, Y extends to an entire function of exponential type ≤ 1 . Non-negativity of the spectral measure $|d\Phi|^2$ together with Akhiezer’s theorem places Y in the Laguerre–Pólya class, so all complex zeros of Y are real. On $\mathbb{R}$ the identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is the rearrangement of (1), with $\sqrt{\Theta'(t)} > 0$. This unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$ is a bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$ with preservation of multiplicity. The critical-line identity $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ for $t \in \mathbb{R}$, combined with reality of zeros of Y and the unitary transfer, fixes the zeros of ζ on the critical line as the Θ^{-1} -image of the real zeros of Y .

1 Introduction

The Hardy Z-function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real-valued on $\mathbb{R}$. The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$ ties the zeros of Z on $\mathbb{R}$ to the zeros of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$ under $s = \frac{1}{2} + it$: the map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$, and $|e^{i\theta(t)}| = 1$ for $t \in \mathbb{R}$ makes the two sides vanish simultaneously.

The Riemann–Siegel phase θ fails to be monotone on $[0, \infty)$: θ' is strictly increasing on $[0, \infty)$ but negative on $[0, t_c]$ and positive on $[t_c, \infty)$, where $t_c \approx 6.2898$ is the unique real zero of θ' . Monotonicity on all of $\mathbb{R}$ is restored by the linear shift $\Theta(t) := \theta(t) + ct$ with $c > c_* := -\theta'(0)$, which enforces $\Theta' > 0$ on $\mathbb{R}$ and makes $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ a C^∞ bijection. The unitary inverse phase pullback $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ is defined in terms of this real bijection.

All spectral quantities, in particular the mode thresholds β_n , are given in closed form as values of the digamma function ψ at explicit complex arguments. Asymptotic expansions are derived from these exact forms and used only where needed for the finiteness of the exceptional set $\mathcal{S}(\mu)$.

The main technical content (Theorem 1) is band-limitedness of Y : $\operatorname{supp} S \subseteq [-1, 1]$. Paley–Wiener then gives Y entire of exponential type ≤ 1 ; Akhiezer gives reality of its complex zeros.

The real-line identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is a unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$, and so is a zero-set bijection with multiplicity preservation between Y on $\mathbb{R}$ and Z on $\mathbb{R}$ (Theorem 3). The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ on $t \in \mathbb{R}$ yields the conclusion for the zeros of ζ on the critical line.

2 The Phase Function

For $t \in \mathbb{C}$ with $\frac{1}{4} + \frac{it}{2} \notin \mathbb{Z}_{\leq 0}$,

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad (2)$$

$$\theta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \quad (3)$$

$$\theta''(t) = -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right) \quad (4)$$

For $t \in \mathbb{R}$, $\frac{1}{4} + \frac{it}{2}$ avoids $\mathbb{Z}_{\leq 0}$ (its real part is $\frac{1}{4}$), so $\theta, \theta', \theta''$ are C^∞ on $\mathbb{R}$.

Lemma 1 (Positivity of θ''). *For every $t > 0$, $\theta''(t) > 0$.*

Proof. $\psi'(s) = \sum_{k \geq 0} (s+k)^{-2}$. With $z_k = \frac{1}{4} + k + \frac{it}{2}$,

$$\operatorname{Im} z_k^{-2} = \frac{-2(\frac{1}{4} + k)(t/2)}{|z_k|^4} < 0 \quad (k \geq 0, t > 0) \quad (5)$$

Summing, $\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) < 0$, therefore $\theta''(t) > 0$. □

Lemma 2 (Exact value of $\theta'(0)$).

$$\theta'(0) = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{1}{2} \log \pi = -\frac{1}{2} \left(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right) \quad (6)$$

Proof. $\psi(\frac{1}{4}) = -\gamma - 3 \log 2 - \frac{\pi}{2}$ from $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$ (reflection at $x = \frac{1}{4}$) and $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6 \log 2$ ([1], §6.3.4). □

Definition 1 (Admissible constant).

$$c_* := -\theta'(0) = \frac{1}{2} \left(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right). \quad (7)$$

A real c is admissible if $c > c_*$. Throughout the paper c is admissible.

Lemma 3 (Monotone bijection). *For every admissible c , $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$, and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing C^∞ bijection.*

Proof. θ' is even on $\mathbb{R}$ (from (3); $\operatorname{Re} \psi(\frac{1}{4} + \frac{i\tau}{2})$ is even in τ) and by Lemma 1 $\theta'' > 0$ on $(0, \infty)$, so θ' is strictly increasing on $[0, \infty)$ and attains its minimum on $\mathbb{R}$ at $\tau = 0$, namely $\theta'(0) = -c_*$. For any $c > c_*$, $\Theta'(\tau) = \theta'(\tau) + c \geq -c_* + c > 0$ on $\mathbb{R}$. The standard asymptotic $\theta(\tau) = \frac{\tau}{2} \log(\tau/(2\pi)) - \frac{\tau}{2} + O(1)$ as $\tau \rightarrow \infty$ gives $\Theta(\tau) \rightarrow +\infty$ as $\tau \rightarrow +\infty$ and $\Theta(\tau) \rightarrow -\infty$ as $\tau \rightarrow -\infty$ (the second by the functional relation $\theta(-\tau) = -\theta(\tau)$ on $\mathbb{R}$). Strict monotonicity together with these limits gives that $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection. Smoothness is inherited from $\theta \in C^\infty(\mathbb{R})$. □

3 The Unitary Inverse Phase Pullback

Definition 2 (Unitary inverse phase pullback). *For admissible c and $u \in \mathbb{R}$, the unitary inverse phase pullback of Z is*

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \quad (8)$$

The map Θ^{-1} is the inverse of the bijection $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ from Lemma 3; $\Theta'(\Theta^{-1}(u))$ is a strictly positive real number, and $\sqrt{\Theta'(\Theta^{-1}(u))}$ denotes its principal (positive) square root. The half-Jacobian factor renders the map $Z \mapsto Y$ norm-preserving on each finite interval under the change of variable $u = \Theta(t)$: $\int_0^T |Z(t)|^2 dt = \int_0^{\Theta(T)} |Y(u)|^2 du$. Globally, Y is not in $L^2(\mathbb{R})$: by Hardy–Littlewood, $\int_0^U |Y(u)|^2 du = \int_0^{\Theta^{-1}(U)} |Z(t)|^2 dt = T \log T + O(T)$ with $T = \Theta^{-1}(U)$ and hence $\rightarrow \infty$ as $U \rightarrow \infty$.

Y is real-valued on $\mathbb{R}$ because Z is real-valued on $\mathbb{R}$ and the denominator is a strictly positive real number.

4 Band-Limitedness

Definition 3 (Mode threshold). *For each $n \geq 1$,*

$$\beta_n := \Theta'(2\pi n^2) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) - \frac{1}{2} \log \pi + c \quad (9)$$

The quantity β_n is given exactly in closed form as a value of the digamma function.

Lemma 4 (Asymptotic of β_n from the exact form). *As $n \rightarrow \infty$,*

$$\beta_n = \log n + c + O(n^{-4}) \quad (10)$$

Proof. The digamma asymptotic

$$\psi(s) = \log s - \frac{1}{2s} - \sum_{k \geq 1} \frac{B_{2k}}{2k s^{2k}}, \quad |\arg s| < \pi \quad (11)$$

applied at $s = \frac{1}{4} + i\pi n^2$ with $|s| = \sqrt{\frac{1}{16} + \pi^2 n^4}$ yields

$$\operatorname{Re} \psi(s) = \log |s| + O(|s|^{-2}) = \frac{1}{2} \log\left(\frac{1}{16} + \pi^2 n^4\right) + O(n^{-4}) = 2 \log n + \log \pi + O(n^{-4}) \quad (12)$$

Therefore $\frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) = \log n + \frac{1}{2} \log \pi + O(n^{-4})$ and $\beta_n = \log n + c + O(n^{-4})$. $\square$

Definition 4 (Truncated transform and power spectral density). *For $T > 0$, $\mu \in \mathbb{R}$,*

$$K_T(\mu) := \frac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du. \quad (13)$$

The power spectral density of Y at frequency μ is

$$S(\mu) := \lim_{T \rightarrow \infty} \frac{2\pi |K_T(\mu)|^2}{\Theta(T)} \quad (14)$$

whenever the limit exists. Equivalently, S is the Fourier transform of the stationary covariance

$$C(\eta) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U Y(u) \overline{Y(u+\eta)} du, \quad (15)$$

when Y has a stationary covariance in the sense of Wiener.

Lemma 5 (Riemann–Siegel identity). *For $t > 0$,*

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\theta(t) - t \log n) + R(t), \quad N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R(t) = O(t^{-1/4}) \quad (16)$$

Proof. [5], §7.4, Theorem 7.1. □

Theorem 1 (Band-limitedness of Y). *For every admissible c and every $\mu \in \mathbb{R}$ with $|\mu| > 1$,*

$$S(\mu) = \lim_{T \rightarrow \infty} \frac{2\pi |K_T(\mu)|^2}{\Theta(T)} = 0. \quad (17)$$

Equivalently, the power spectral density of Y is supported in $[-1, 1]$.

Proof. The proof identifies Z as a Clerc–Mallat warped process via the Omer–Torresani unitary warping operator, pulls the windowed integrand through the warping back to the u -variable of the warped target Y , and applies a single integration by parts in u per mode.

(A) *Clerc–Mallat identification.* Write $\mathcal{D}_\Theta Y$ for the Omer–Torresani warping operator $[\mathcal{D}_\Theta Y](t) := \sqrt{\Theta'(t)} Y(\Theta(t))$. By (8), $Z = \mathcal{D}_\Theta Y$ on $\mathbb{R}$. The conditions $\Theta \in C^\infty(\mathbb{R})$ and $\Theta' > 0$ on $\mathbb{R}$ (Lemma 3) place Z in the Clerc–Mallat deformed stationary class in the original sense of CM-Eq. 3 [2], and in the Omer–Torresani unitary form without the upper-bound condition $\Theta' \leq C < \infty$; the unboundedness of Θ' at infinity is benign for unitarity. In particular $\mathcal{D}_\Theta$ is a unitary bijection $L_{\text{loc}}^2(\mathbb{R}) \rightarrow L_{\text{loc}}^2(\mathbb{R})$ under the change of variable $u = \Theta(t)$, and the identity (13) expresses $K_T(\mu)$ as a pure Fourier integral of Y in the u -variable:

$$K_T(\mu) = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du. \quad (18)$$

All subsequent analysis is in the u -variable.

(B) *Riemann–Siegel factorization of Y .* Substituting (16) into (8) and expanding the cosine, for u in the image $\Theta([0, \infty))$, with $t = \Theta^{-1}(u)$,

$$Y(u) = \sum_{\sigma \in \{\pm 1\}} \sum_{n=1}^{N(t(u))} n^{-1/2} Y_{n,\sigma}(u) + Y_R(u), \quad (19)$$

$$Y_{n,\sigma}(u) := \frac{1}{\sqrt{\Theta'(t(u))}} e^{i\sigma[\theta(t(u)) - t(u) \log n]}, \quad Y_R(u) := \frac{R(t(u))}{\sqrt{\Theta'(t(u))}}. \quad (20)$$

Let $u_n := \Theta(2\pi n^2)$; mode (n, σ) contributes to (19) on $\{u \geq u_n\}$. Substituting (19) into (18) gives

$$K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma \in \{\pm 1\}} \sum_{n \leq N(T)} n^{-1/2} \mathcal{J}_{n,\sigma}(T, \mu) + \mathcal{R}(T, \mu), \quad (21)$$

with the mode integrals and remainder integral taken in the u -variable

$$\mathcal{J}_{n,\sigma}(T, \mu) := \int_{u_n}^{\Theta(T)} G_{n,\sigma}(u) e^{i\Psi_{n,\sigma,\mu}(u)} du, \quad G_{n,\sigma}(u) := \frac{1}{\sqrt{\Theta'(t(u))}}, \quad (22)$$

$$\Psi_{n,\sigma,\mu}(u) := \sigma[\theta(t(u)) - t(u) \log n] - \mu u, \quad (23)$$

$$\mathcal{R}(T, \mu) := \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y_R(u) e^{-i\mu u} du. \quad (24)$$

(C) *Instantaneous u -frequency of $Y_{n,\sigma}$.* The phase of $Y_{n,\sigma}$ is $\Psi_{n,\sigma}^0(u) := \sigma[\theta(t(u)) - t(u) \log n]$. Using $dt/du = 1/\Theta'(t) = 1/(\theta'(t) + c)$,

$$\frac{d\Psi_{n,\sigma}^0}{du}(u) = \sigma \frac{\theta'(t(u)) - \log n}{\theta'(t(u)) + c}. \quad (25)$$

On the support $\{u \geq u_n\}$ of mode (n, σ) , strict monotonicity of θ' (Lemma 1) gives $\theta'(t(u)) \geq \theta'(2\pi n^2) = \beta_n - c = \log n + r_n$ with $r_n = O(n^{-4})$ by Lemma 4. Hence the numerator $\theta'(t) - \log n \geq r_n$ and the denominator $\theta'(t) + c \geq \beta_n = \log n + c + r_n \geq c - c_* > 0$. The numerator is less than the denominator by exactly the positive quantity c , and the denominator is bounded below by $c - c_* > 0$ while $r_n \rightarrow 0$; consequently

$$\left| \frac{d\Psi_{n,\sigma}^0}{du}(u) \right| < 1 \quad \text{for every } u \geq u_n \text{ and every } n \geq 1. \quad (26)$$

Explicitly, $\Psi_{n,+}^{0'}$ lies in $[0, 1)$ and $\Psi_{n,-}^{0'}$ in $(-1, 0]$.

(D) *Per-mode IBP for $|\mu| > 1$.* For $|\mu| > 1$, (26) gives the uniform phase-derivative lower bound

$$|\Psi'_{n,\sigma,\mu}(u)| = \left| \frac{d\Psi_{n,\sigma}^0}{du}(u) - \mu \right| \geq |\mu| - 1 =: \lambda > 0, \quad u \geq u_n, \quad (27)$$

for every (n, σ) ; the sign of $\Psi'_{n,\sigma,\mu}(u)$ is constant on $[u_n, \infty)$ because the inequality (26) prevents $\Psi'_{n,\sigma,\mu}$ from crossing zero. One integration by parts on $\mathcal{J}_{n,\sigma}$ with $G_{n,\sigma}$ against $e^{i\Psi_{n,\sigma,\mu}}$,

$$\mathcal{J}_{n,\sigma}(T, \mu) = \left[\frac{G_{n,\sigma}(u) e^{i\Psi_{n,\sigma,\mu}(u)}}{i\Psi'_{n,\sigma,\mu}(u)} \right]_{u_n}^{\Theta(T)} - \int_{u_n}^{\Theta(T)} \frac{d}{du} \left(\frac{G_{n,\sigma}(u)}{i\Psi'_{n,\sigma,\mu}(u)} \right) e^{i\Psi_{n,\sigma,\mu}(u)} du. \quad (28)$$

The boundary term is bounded by $2 \sup_{u \geq u_n} |G_{n,\sigma}(u)|/\lambda \leq 2/(\lambda\sqrt{c - c_*})$, independent of T . For the tail,

$$G'_{n,\sigma}(u) = -\frac{\theta''(t(u))}{2\Theta'(t(u))^{5/2}},$$

and the substitution $u = \Theta(t)$ gives

$$\begin{aligned} \int_{u_n}^{\infty} |G'_{n,\sigma}(u)| du &= \int_{2\pi n^2}^{\infty} \frac{\theta''(t)}{2\Theta'(t)^{5/2}} \Theta'(t) dt = \int_{2\pi n^2}^{\infty} \frac{\theta''(t)}{2\Theta'(t)^{3/2}} dt \\ &= \left[-\Theta'(t)^{-1/2} \right]_{2\pi n^2}^{\infty} = \beta_n^{-1/2}, \end{aligned}$$

using $\theta'' > 0$ (Lemma 1) and $\Theta'(t) \rightarrow \infty$ as $t \rightarrow \infty$. The second-phase-derivative estimate

$$|\Psi''_{n,\sigma,\mu}(u)| = \left| \frac{d}{du} \frac{d\Psi_{n,\sigma}^0}{du} \right| = \frac{c\theta''(t(u))}{\Theta'(t(u))^3} \leq \frac{c\theta''(t(u))}{\Theta'(t(u))^2}$$

gives

$$\int_{u_n}^{\infty} \frac{|\Psi''_{n,\sigma,\mu}(u)|}{\Theta'(t(u))^{-1/2} \vee 1} du \leq c \int_{2\pi n^2}^{\infty} \frac{\theta''(t)}{\Theta'(t)^{3/2}} dt = O(\beta_n^{-1/2}).$$

Therefore the tail integrand in (28) is dominated by $|G'_{n,\sigma}|/\lambda + |G_{n,\sigma}||\Psi''_{n,\sigma,\mu}|/\lambda^2$, and its integral is $O(\lambda^{-2}\beta_n^{-1/2})$ uniformly in T . Consequently

$$|\mathcal{J}_{n,\sigma}(T, \mu)| \leq \frac{C_0(c)}{\lambda} + \frac{C_1(c)}{\lambda^2 \sqrt{\beta_n}} \leq \frac{C_2(c)}{\lambda} \quad (29)$$

for every T and every (n, σ) .

(E) *Remainder contribution.* The Riemann–Siegel remainder $R(t)$ admits the Riemann–Siegel contour-integral representation ([5], §7.4) whose t -derivative of the argument is $O((t/(2\pi))^{-1/2})$; pulled back, $Y_R(u)$ has instantaneous u -frequency $O((t(u)/(2\pi))^{-1/2})/\Theta'(t(u)) \rightarrow 0$, hence lies in $(-\frac{1}{2}, \frac{1}{2})$ for u sufficiently large. Pulled-back amplitude satisfies $|Y_R(u)| = |R(t(u))|/\sqrt{\Theta'(t(u))} = O(t(u)^{-1/4}/\sqrt{\log t(u)})$, which is integrable against $e^{-i\mu u}$ after one integration by parts with phase derivative $|- \mu + O(\cdot)| \geq \frac{1}{2}(|\mu| - 1) > 0$ for u large. Therefore

$$|\mathcal{R}(T, \mu)| \leq C_R(|\mu|) \quad \text{uniformly in } T. \quad (30)$$

(F) *Power spectral density vanishing.* Combining (29) and (30) in (21),

$$|K_T(\mu)| \leq \frac{C_2(c)}{2\pi\lambda} \sum_{\sigma} \sum_{n \leq N(T)} n^{-1/2} + C_R(|\mu|) = O(N(T)^{1/2}) = O(T^{1/4}), \quad (31)$$

using $\sum_{n \leq N} n^{-1/2} = O(N^{1/2})$ and $N(T) = \lfloor \sqrt{T/(2\pi)} \rfloor$. Since $\Theta(T) \sim \frac{T}{2} \log T$,

$$\frac{2\pi |K_T(\mu)|^2}{\Theta(T)} = O\left(\frac{T^{1/2}}{T \log T}\right) = O((T \log T)^{-1/2}) \rightarrow 0. \quad (32)$$

Therefore $S(\mu) = 0$ for every μ with $|\mu| > 1$.

(G) *Frequency-location summary.* The $\sigma = +1$ modes have instantaneous u -frequency in $[0, 1)$, the $\sigma = -1$ modes in $(-1, 0]$ (part (C)), and the Riemann–Siegel remainder contribution has instantaneous u -frequency tending to $0 \in (-1, 1)$ (part (E)). This is the frequency-location data used by Corollary 1. □

Corollary 1 (Spectral measure from the inverse integral). *For $U > 0$ and $\xi \in \mathbb{R}$, define the one-sided windowed inverse integral*

$$\Phi_U^+(\xi) := \frac{1}{2\pi} \int_0^U Y(u) e^{-i\xi u} du. \quad (33)$$

The mean-square limit

$$\Phi^+(\xi) := \lim_{U \rightarrow \infty} \Phi_U^+(\xi) \quad (34)$$

exists in $L^2([0, 1], d\xi)$. Setting

$$\Phi(\xi) := \begin{cases} \Phi^+(\xi) & 0 \leq \xi \leq 1, \\ \overline{\Phi^+(-\xi)} & -1 \leq \xi < 0, \\ 0 & |\xi| > 1, \end{cases} \quad (35)$$

defines a function $\Phi \in L^2([-1, 1], d\xi)$. The associated complex Borel measure $d\Phi(\xi) := \Phi(\xi) d\xi$ satisfies the Hermitian symmetry $d\Phi(-\xi) = \overline{d\Phi(\xi)}$ and gives the Cramér representation

$$Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi), \quad u \in \mathbb{R}. \quad (36)$$

Proof. Convergence on $\xi \in (0, 1)$. Substitute $u = \Theta(t)$, $du = \Theta'(t) dt$, $Y(\Theta(t)) = Z(t)/\sqrt{\Theta'(t)}$; then

$$\Phi_U^+(\xi) = \frac{1}{2\pi} \int_{\Theta^{-1}(\Theta(0) \vee 0)}^{\Theta^{-1}(U)} Z(t) \sqrt{\Theta'(t)} e^{-i\xi\Theta(t)} dt, \quad (37)$$

and inserting the Riemann–Siegel expansion (16), $\Phi_U^+(\xi)$ is a finite sum of mode integrals with phase derivative $\Phi'_{n,\sigma,\xi}(t) = (\sigma - \xi)\Theta'(t) - \sigma(\log n + c)$ identical to the phase analysis of Theorem 1, plus a remainder with phase derivative $-\xi\Theta'(t)$ treated as in step (i). For $\xi \in (0, 1)$, each mode (n, σ) has phase slope tending to $\pm\infty$ with constant sign past a bounded threshold, so the integration-by-parts analysis of steps (iii)–(vi) applies uniformly on compact subsets of $(0, 1)$ and produces the bound

$$|\Phi_U^+(\xi) - \Phi_{U'}^+(\xi)| = O((\log \Theta^{-1}(U \wedge U'))^{-1/2}) \quad (38)$$

uniformly on each compact $[\xi_0, 1 - \xi_0] \subset (0, 1)$. Hence $\{\Phi_U^+\}_{U>0}$ is Cauchy in $L^2_{\text{loc}}([0, 1], d\xi)$. Parseval on the windowed inverse integral,

$$\int_{\mathbb{R}} |\Phi_U^+(\xi)|^2 d\xi = \frac{1}{2\pi} \int_0^U |Y(u)|^2 du = \frac{1}{2\pi} \int_0^{\Theta^{-1}(U)} |Z(t)|^2 dt, \quad (39)$$

combined with Theorem 1 which gives $\Phi_U^+(\xi) \rightarrow 0$ in L^2 -sense for $|\xi| > 1$, concentrates the L^2 mass of Φ_U^+ on $[-1, 1]$. The mean-square limit $\Phi^+ \in L^2([0, 1], d\xi)$ therefore exists.

Divergence at $\xi = 0$ for the $\sigma = -1$ modes. At $\xi = 0$, the $\sigma = -1$ mode has phase slope $-\Theta'(t) + (\log n + c) \rightarrow -\infty$ with amplitude $\sqrt{\Theta'(t)}$; integration by parts converges. But at $\xi = 0$ the $\sigma = +1$ mode has slope $\Theta'(t) - (\log n + c)$, which vanishes at the stationary point t_n^* defined by $\Theta'(t_n^*) = \log n + c$, and the integral has a stationary-phase contribution $O(\sqrt{t_n^*})$ that does not decay. Summed over $n \leq N(T)$ this is $O(T^{1/2})$, so $\Phi_U^+(0)$ is unbounded as $U \rightarrow \infty$: the mean-square limit does not extend continuously to $\xi = 0$ from the positive side. The reflection defines Φ on $[-1, 0]$ by (35).

Hermitian symmetry. Y is real on $\mathbb{R}$, so $\overline{\Phi_U^+(\xi)} = \frac{1}{2\pi} \int_0^U Y(u) e^{i\xi u} du$, which is $\Phi_U^+(-\xi)$ with argument reflected; passing to the mean-square limit, $\overline{\Phi^+(\xi)}$ is the mean-square limit of the inverse integral evaluated at $-\xi$ on the same half-line. The definition (35) makes this Hermitian extension exact: $\overline{\Phi(-\xi)} = \Phi(\xi)$ for $\xi \in [-1, 1]$.

Representation (36). For $u \in \mathbb{R}$, Plancherel applied to (34) gives

$$Y(u) = \int_{-1}^1 e^{i\xi u} \Phi(\xi) d\xi = \int_{-1}^1 e^{i\xi u} d\Phi(\xi), \quad (40)$$

the L^2 -Fourier inversion theorem applied to the function $\Phi \in L^2([-1, 1], d\xi) \subset L^2(\mathbb{R}, d\xi)$, whose Fourier transform is Y on $\mathbb{R}$ by Theorem 1 and (39). $\square$

Proposition 1 (Per-mode factorization of Φ). *For each $n \geq 1$ and $\sigma \in \{\pm 1\}$, let*

$$t_{n,\sigma}^*(\xi) := (\Theta')^{-1} \left(\frac{\log n + c}{1 - \sigma\xi} \right), \quad (41)$$

defined for those ξ with $1 - \sigma\xi > 0$ and $(\log n + c)/(1 - \sigma\xi) \geq \beta_n$; equivalently (using (10) with $r_n = O(n^{-4})$), for

$$\xi \in J_{n,+} := [r_n/\beta_n, 1) \quad (\sigma = +1), \quad \xi \in J_{n,-} := (-1, -r_n/\beta_n] \quad (\sigma = -1). \quad (42)$$

Let

$$|\mathcal{A}_{n,\sigma}(\xi)|^2 := \frac{2\pi(\log n + c)}{(1 - \sigma\xi)(\sigma - \xi)^2 \theta''(t_{n,\sigma}^*(\xi))}, \quad (43)$$

and

$$\varphi_{n,\sigma}(\xi) := \sigma\theta(t_{n,\sigma}^*(\xi)) - \sigma t_{n,\sigma}^*(\xi) \log n - \xi \Theta(t_{n,\sigma}^*(\xi)) + \frac{\pi}{4}\sigma. \quad (44)$$

Then the density Φ from Corollary 1 admits the mode-by-mode factorization

$$\Phi(\xi) = \frac{1}{2\pi} \sum_{\sigma \in \{\pm 1\}} \sum_{n=1}^{\infty} n^{-1/2} \mathcal{A}_{n,\sigma}(\xi) e^{i\varphi_{n,\sigma}(\xi)} \mathbf{1}_{J_{n,\sigma}}(\xi) + \Phi_R(\xi), \quad \xi \in [-1, 1], \quad (45)$$

where Φ_R is the Riemann–Siegel remainder contribution obtained by inserting $R(t)$ into Φ_U^+ and passing to the mean-square limit. The amplitude $|\mathcal{A}_{n,\sigma}(\xi)|^2$ is integrable on $J_{n,\sigma}$ in the sense that

$$\int_{J_{n,\sigma}} |\mathcal{A}_{n,\sigma}(\xi)|^2 d\xi = \int_{r_n/\beta_n}^1 \frac{2\pi(\log n + c)}{v^3 \theta''((\Theta')^{-1}((\log n + c)/v))} dv \quad (46)$$

under the substitution $v := 1 - \sigma\xi$ (so $v \in [r_n/\beta_n, 1]$, $\xi = \sigma(1 - v)$, $(1 - \sigma\xi) = v$, $(\sigma - \xi)^2 = v^2$). The lower endpoint $v = r_n/\beta_n > 0$ is strictly positive by (10), so the integrand $2\pi(\log n + c)/[v^3 \theta''(t^*(v))]$ is continuous and bounded on $[r_n/\beta_n, 1]$ and the integral is finite. Summing (46) over (n, σ) together with $\|\Phi_R\|_{L^2}^2$ equals $\|\Phi\|_{L^2([-1,1])}^2$, which by (39) is the mean-square limit of $(2\pi)^{-1} \int_0^U |Y(u)|^2 du$.

Proof. Substitute $u = \Theta(t)$, $du = \Theta'(t) dt$, $Y(\Theta(t)) = Z(t)/\sqrt{\Theta'(t)}$ into $\Phi_U^+(\xi)$ as in (37); insert the Riemann–Siegel expansion (16) and expand $\cos \alpha = \frac{1}{2}(e^{i\alpha} + e^{-i\alpha})$, giving

$$\Phi_U^+(\xi) = \frac{1}{2\pi} \sum_{\sigma} \sum_{n \leq N(\Theta^{-1}(U))} n^{-1/2} \int_{t_n}^{\Theta^{-1}(U)} \sqrt{\Theta'(t)} e^{i\Psi_{n,\sigma,\xi}(t)} dt + \Phi_{U,R}^+(\xi), \quad (47)$$

with $\Psi_{n,\sigma,\xi}(t) := \sigma\theta(t) - \sigma t \log n - \xi\Theta(t)$, so $\Psi'_{n,\sigma,\xi}(t) = (\sigma - \xi)\Theta'(t) - \sigma(\log n + c)$ and $\Psi''_{n,\sigma,\xi}(t) = (\sigma - \xi)\theta''(t)$ (since $\Theta'' = \theta''$). The stationary point $t_{n,\sigma}^*(\xi)$ satisfies $\Psi'_{n,\sigma,\xi}(t^*) = 0$, i.e. $\Theta'(t^*) = (\log n + c)/(1 - \sigma\xi)$, which inverts to (41) on $J_{n,\sigma}$. For ξ in the interior of $J_{n,\sigma}$ the stationary point lies in (t_n, ∞) with $\Psi''_{n,\sigma,\xi}(t^*) = (\sigma - \xi)\theta''(t^*) \neq 0$, and classical stationary phase ([6], Theorem 7.7.5) produces

$$\int_{t_n}^{\Theta^{-1}(U)} \sqrt{\Theta'(t)} e^{i\Psi_{n,\sigma,\xi}(t)} dt = \sqrt{\frac{2\pi \Theta'(t^*)}{|\Psi''_{n,\sigma,\xi}(t^*)|}} e^{i\Psi_{n,\sigma,\xi}(t^*) + i(\pi/4) \operatorname{sgn} \Psi''_{n,\sigma,\xi}(t^*)} + O(\Theta'(t^*)^{-1/2}),$$

uniformly for ξ in any compact subset of the interior of $J_{n,\sigma}$ and $U \geq t^*$. Using $|\sigma - \xi| = 1 - \sigma\xi$ on $J_{n,\sigma}$ and $\operatorname{sgn}(\sigma - \xi) = \sigma$, the leading factor is exactly $\mathcal{A}_{n,\sigma}(\xi) e^{i\varphi_{n,\sigma}(\xi)}$ in (43)–(44). The $O(\Theta'(t^*)^{-1/2})$ remainder vanishes in the mean-square limit on compact subsets of the interior of $J_{n,\sigma}$ by the IBP tail bound (38). Passing to $U \rightarrow \infty$ in (47) in L^2 gives (45).

For (46): substitute $v = 1 - \sigma\xi$, so $\xi = \sigma(1 - v)$, $d\xi = -\sigma dv$, $(\sigma - \xi) = \sigma v$ hence $(\sigma - \xi)^2 = v^2$, $(1 - \sigma\xi) = v$; the range $\xi \in J_{n,\sigma}$ becomes $v \in [r_n/\beta_n, 1]$. Substituting into (43) gives the integrand $2\pi(\log n + c)/[v^3 \theta''(t^*(v))]$ as stated. Summability: $\int |\Phi|^2 d\xi$ is finite by (39); non-negativity of $|\mathcal{A}_{n,\sigma}(\xi)|^2 \mathbf{1}_{J_{n,\sigma}}(\xi) + |\Phi_R(\xi)|^2$ and Tonelli's theorem reduce the (n, σ) sum and the ξ -integral to a single integration of a non-negative function, whose total equals $\|\Phi\|_{L^2}^2$. $\square$

5 Entire Extension and Laguerre–Pólya Membership

Corollary 2 (Entire extension of exponential type ≤ 1). *Y extends to an entire function of exponential type ≤ 1 :*

$$Y(z) = \int_{-1}^1 e^{i\xi z} d\Phi(\xi), \quad |Y(z)| \leq \|d\Phi\| e^{|\operatorname{Im} z|}, \quad (48)$$

where $d\Phi$ is the Cramér spectral measure from (36) and $\|d\Phi\|$ is its total variation.

Proof. Define $Y(z) := \int_{-1}^1 e^{i\xi z} d\Phi(\xi)$ for $z \in \mathbb{C}$. The integrand is entire in z for each fixed $\xi \in [-1, 1]$; the measure $d\Phi$ is finite (total variation $\|d\Phi\| < \infty$) and the integrand is uniformly bounded on compact subsets of $\mathbb{C}$ by e^R for $|z| \leq R$ since $|\xi| \leq 1$. By the dominated convergence theorem and Morera's theorem, Y is entire. For the exponential-type bound,

$$|Y(z)| \leq \int_{-1}^1 |e^{i\xi z}| |d\Phi|(\xi) = \int_{-1}^1 e^{-\xi \operatorname{Im} z} |d\Phi|(\xi) \leq e^{|\operatorname{Im} z|} \int_{-1}^1 |d\Phi|(\xi) = \|d\Phi\| e^{|\operatorname{Im} z|}. \quad (49)$$

Restricted to $z = u \in \mathbb{R}$, (48) reproduces (36); uniqueness of the entire extension of the real-analytic function $u \mapsto Y(u)$ (with globally convergent representation) identifies this extension with Y on $\mathbb{R}$. $\square$

Lemma 6 (Non-negativity of the spectral measure). *The spectral measure $|d\Phi|^2$ of the representation (36) is a non-negative finite Borel measure on $[-1, 1]$.*

Proof. From (36), the Cramér covariance of Y is $C(\eta) = \int_{-1}^1 e^{i\xi\eta} d|\Phi|^2(\xi)$ with $|d\Phi|^2$ non-negative and supported in $[-1, 1]$ by construction. The power spectral density S from (14) equals the Radon–Nikodym derivative of the absolutely continuous part of $|d\Phi|^2$; Theorem 1 gives $S(\mu) = 0$ for $|\mu| > 1$, consistent with the support containment. $\square$

Theorem 2 (Akhiezer). *A real-valued entire function F of exponential type $\leq \tau$ whose spectral measure $|\hat{F}|^2$ is a non-negative measure supported in $[-\tau, \tau]$ belongs to the Laguerre–Pólya class. All zeros of F in $\mathbb{C}$ are real.*

Proof. [7], Lecture 16, Theorem 3; [4], §V.4. $\square$

Corollary 3 (Reality of zeros of Y). *All zeros of Y in $\mathbb{C}$ are real.*

Proof. Y is real on $\mathbb{R}$ (Definition 2), entire of exponential type ≤ 1 (Corollary 2), with spectral measure $|d\Phi|^2$ non-negative and supported in $[-1, 1]$ (Lemma 6). Apply Theorem 2. $\square$

6 Transfer to the Hardy Z-Function on the Real Line

The transfer uses only the real-line identity

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)}, \quad t \in \mathbb{R}, \quad (50)$$

which is the rearrangement of (8) after substituting $u = \Theta(t)$. On $\mathbb{R}$, $\Theta'(t) > 0$ (Lemma 3) and $\sqrt{\Theta'(t)}$ is the principal (positive) square root of a strictly positive real number; no branch choice is involved.

Theorem 3 (Zero-set bijection under the unitary time change on $\mathbb{R}$). *Let c be admissible. The map $t \mapsto \Theta(t)$ is a bijection $\mathbb{R} \rightarrow \mathbb{R}$ that restricts to a bijection of zero sets,*

$$\{t \in \mathbb{R} : Z(t) = 0\} \longleftrightarrow \{u \in \mathbb{R} : Y(u) = 0\}, \quad u = \Theta(t), \quad (51)$$

with preservation of multiplicity: if $t_0 \in \mathbb{R}$ and Y has a zero of order m at $\Theta(t_0)$, then Z has a zero of order exactly m at t_0 , and conversely.

Proof. $\sqrt{\Theta'(t)}$ is strictly positive on $\mathbb{R}$. From (50), for any $t_0 \in \mathbb{R}$,

$$Z(t_0) = 0 \iff \sqrt{\Theta'(t_0)} Y(\Theta(t_0)) = 0 \iff Y(\Theta(t_0)) = 0. \quad (52)$$

Since $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection (Lemma 3), $t_0 \mapsto \Theta(t_0)$ is a bijection between the real zero set of Z and the real zero set of Y .

For multiplicity, let $t_0 \in \mathbb{R}$ and suppose Y has a zero of order m at $s_0 := \Theta(t_0)$. Since Y is entire (Corollary 2), there exists an entire g with $g(s_0) \neq 0$ and

$$Y(s) = (s - s_0)^m g(s). \quad (53)$$

Substitute $s = \Theta(t)$ (defined and C^∞ for $t \in \mathbb{R}$):

$$Y(\Theta(t)) = (\Theta(t) - \Theta(t_0))^m g(\Theta(t)). \quad (54)$$

The Hadamard-type identity

$$\Theta(t) - \Theta(t_0) = (t - t_0) h(t), \quad h(t) := \int_0^1 \Theta'(t_0 + u(t - t_0)) du, \quad (55)$$

valid for all $t \in \mathbb{R}$ because $\Theta \in C^1(\mathbb{R})$, gives $h(t_0) = \Theta'(t_0) > 0$, and continuity of h together with $\Theta' > 0$ on $\mathbb{R}$ ensures $h(t) > 0$ for t in a neighborhood of t_0 . Combining (50), (54), and (55),

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)} = (t - t_0)^m h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (56)$$

Denote the trailing factor

$$H(t) := h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (57)$$

Then H is continuous at t_0 and

$$H(t_0) = \Theta'(t_0)^m g(s_0) \sqrt{\Theta'(t_0)} \neq 0 \quad (58)$$

because $\Theta'(t_0) > 0$ and $g(s_0) \neq 0$. Therefore Z has a zero of order exactly m at t_0 .

The converse (given a zero of Z of order m at $t_0 \in \mathbb{R}$, Y has a zero of order m at $\Theta(t_0)$) is obtained by applying the same argument with Θ replaced by Θ^{-1} , which satisfies $(\Theta^{-1})'(u) = 1/\Theta'(\Theta^{-1}(u)) > 0$ for $u \in \mathbb{R}$ by Lemma 3 and the inverse function theorem. $\square$

Corollary 4 (Zeros of Z on $\mathbb{R}$). *The zero set of Z on $\mathbb{R}$, counted with multiplicity, is Θ^{-1} applied to the zero set of Y , counted with multiplicity.*

Proof. Corollary 3 gives that every zero of the entire function Y lies on $\mathbb{R}$. Theorem 3 gives a multiplicity-preserving bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$, implemented by $t \mapsto \Theta(t)$. $\square$

7 The Zero Locus of ζ on the Critical Line

Corollary 5 (Zeros of ζ on the critical line). *The zero set of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$, counted with multiplicity, is the image of $\Theta^{-1}(\{u \in \mathbb{R} : Y(u) = 0\})$ under the map $t \mapsto \frac{1}{2} + it$, with multiplicities inherited from those of Y .*

Proof. For $t \in \mathbb{R}$, $|e^{i\theta(t)}| = 1$, so $e^{i\theta(t)}$ is a nonzero complex number; hence $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ gives $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$, with matching orders of vanishing at every $t \in \mathbb{R}$ (multiplication by the non-vanishing real-analytic factor $e^{-i\theta(t)}$ does not alter the order of a zero of the real-analytic factor $Z(t)$ at a real point). Corollary 4 identifies the real zero set of Z with Θ^{-1} applied to the zero set of Y . The map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$. Composing gives the stated identification. $\square$

Corollary 6 (Spectral reconstruction of ζ on the critical line). *For every $t \in \mathbb{R}$,*

$$\zeta(\tfrac{1}{2} + it) = e^{-i\theta(t)} \sqrt{\Theta'(t)} \int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi), \quad (59)$$

where $d\Phi$ is the spectral measure from Corollary 1 with the per-mode factorization (45).

Proof. $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ on $\mathbb{R}$ (Introduction). $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ on $\mathbb{R}$ by (50). $Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi)$ for $u \in \mathbb{R}$ by (36). Compose with $u = \Theta(t)$. $\square$

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